



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing–I part–Thursday, February 1st, 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please discuss the following sentence: "In a PRAM model, processors have private local memories". True/False? (1)

- B. Please discuss PRAM prefix sum algorithm according to the Read/Write accesses. (1)

- C. Please discuss the following sentence: "Amdahl's model is classified as a strong scaling model while Gustafson as a weak scaling model". True/False? Why? (2)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Briefly describe what "Divergent" execution" means. (1)

- B. Please discuss the following sentence: "The latency of the memory operation is changed by multi-threading." (1)

- C. Explain when an application is classified as "Bandwidth bound". (1)

- D. Please compare Coarse-grain multithreading with Simultaneous multithreading. (1)

Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. May CUDA kernels create dependencies between threads in a block? (1)

B. Why CUDA kernels cannot exploit context switching? (1)

C. Is there a relation between the number of CUDA threads in a warp and the number of registers in a streaming multi-processor (SM) unit? What else influences the number of registers? (2)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please explain what is meant by the "Corner Turning" technique. (1) **This question can be skipped in case of a passed challenge.**

- B. We say that coalesced access happens if the index in an array is in the form of (fill the dots) (1)

- C. Please define what a memory burst is in terms of DRAM architecture. (1)

- D. Please define what is meant by "Memory consistency." (1)
